

**Project: Rocketry Training: Aerospace
Simulation Project Report (Fem + Cfd)**

**Project: Finite Element Method (Fem)
Structural Stress & Computational
Fluid Dynamics (Cfd) Aerodynamic
Analysis Of A Supersonic Rocket Fin**



INDIA SPACE LAB

Project Report

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CHAPTER 1: AIM

- To perform a Finite Element Method (FEM) structural stress and deformation analysis on a supersonic rocket fin geometry under aerodynamic pressure loading.
- To carry out a Computational Fluid Dynamics (CFD) aerodynamic flow simulation around the fin to determine pressure distribution, velocity contours, and drag coefficient (C_d).
- To execute a Mesh Convergence Study and optimize the fin geometry to reduce peak Von Mises stress and aerodynamic drag.



CHAPTER 2: THEORY (PART A QUESTIONS)

2.1 Finite Element Method (FEM) Overview

The **Finite Element Method (FEM)** is a numerical technique used to find approximate solutions to complex partial differential equations in solid mechanics and fluid flow. Instead of attempting to solve governing equations across a continuous, complex shape simultaneously, FEM discretizes the geometry into smaller, simpler sub-domains called **elements** connected at discrete **nodes**.

- **Why it is used:** Complex geometry, non-linear material behaviors, and varying real-world loading conditions make exact analytical mathematical solutions impossible. FEM converts partial differential equations into algebraic matrix equations $[K]\{u\} = \{F\}$ (where $[K]$ is stiffness, $\{u\}$ is displacement, and $\{F\}$ is force) that computers can solve rapidly.
- **Aerospace Applications:** Structural integrity testing of rocket airframes, wing/fin deflection under aerodynamic drag and lift, thermal stress analysis in propulsion nozzles, and modal vibration analysis to prevent flutter.

2.2 Von Mises Stress & Structural Yielding

Von Mises Stress (σ_{vm}) is a scalar equivalent stress value derived from the 3D principal stress tensor ($\sigma_1, \sigma_2, \sigma_3$). It represents the distortional strain energy accumulated within a solid material:

$$\sigma_{vm} = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}$$

- **Importance in Structural Failure:** Ductile materials (like aerospace aluminum alloys) yield when distortional strain energy exceeds the material's yield strength (σ_y). Von Mises stress provides a single value to compare directly against yield strength to predict permanent plastic deformation and structural failure under multi-axial load conditions.



2.3 Role of Meshing in FEM

Meshing subdivides continuous spatial domain geometry into a finite grid of elements (tetrahedrons, hexahedrons, triangles, or quads).

- **Accuracy vs. Computational Cost:** A fine mesh (small element size) captures sharp stress gradients, geometric fillets, and stress concentrations accurately, but increases node count and computation time exponentially. A coarse mesh computes fast but underpredicts critical stress concentrations.
- **Mesh Convergence:** Engineering simulations perform a **Mesh Convergence Study**. As element size is refined, calculated displacement and stress values converge toward a stable, true value. Once results change by $< 2\%$, mesh independence is established, balancing numerical accuracy with computational cost.

2.4 Differentiating CFD and FEM

Parameter	Computational Fluid Dynamics (CFD)	Finite Element Method (FEM)
Primary Domain	Fluid flow (gases, liquids) around or inside geometry.	Solid mechanics, structural deformation, heat transfer.
Governing Equations	Navier-Stokes Equations (Mass, Momentum, Energy).	Elasticity Equations (Hooke's Law, Strain-Displacement).
Primary Outputs	Velocity vectors, pressure fields, drag (C_d), lift (C_l), turbulence.	Von Mises stress, principal stress, total deformation, Safety Factor.
Aerospace Goal	Optimizing aerodynamic shapes, reducing drag, managing heating.	Ensuring structural strength, preventing structural buckling.



2.5 Importance of Boundary Conditions

Boundary conditions (BCs) define how a computer simulation model interacts with its physical environment by setting known values for displacements, forces, pressure, or velocity at geometric boundaries.

- **Types:** Fixed supports (0 degrees of freedom at mounting points), normal pressures, aerothrust forces, thermal fluxes, and velocity inlets.
- **Impact of Incorrect Assumptions:** If a rocket fin root is modeled as completely rigid when it actually flexes, stress concentrations will be wrongly predicted. Incorrect BCs result in inaccurate, dangerous engineering conclusions.



CHAPTER 3: GEOMETRY

- **Component:** Trapezoidal Supersonic Rocket Fin.
- **Dimensions:**
 - Root Chord (C_r): 150 mm
 - Tip Chord (C_t): 60 mm
 - Semi-Span (b): 120 mm
 - Leading Edge Sweep Angle (Λ): 45°
 - Maximum Thickness (t): 6 mm
- **CFD Domain:** 3D Enclosure domain surrounding fin ($3 \times$ fin span length upstream, $10 \times$ downstream wake zone).



CHAPTER 4: MATERIAL PROPERTIES

- **Selected Material: Aluminum 7075-T6** (Standard High-Strength Aerospace Alloy).
 - Mass Density (ρ): 2810 kg/m³
 - Young's Modulus (E): 71.7 GPa
 - Poisson's Ratio (ν): 0.33
 - Yield Strength (σ_y): 503 MPa
 - Ultimate Tensile Strength (σ_u): 572 MPa



CHAPTER 5: BOUNDARY CONDITIONS

5.1 FEM Structural Boundary Conditions

- **Fixed Support:** Root joint interface attached rigidly to the rocket fuselage casing ($U_x = U_y = U_z = 0, \theta_x = \theta_y = \theta_z = 0$).
- **Aerodynamic Load:** Normal aerodynamic surface pressure applied to windward fin surface during high-speed atmospheric maneuver ($P_{normal} = 15,000 \text{ N/m}^2$).

5.2 CFD Aerodynamic Boundary Conditions

- **Inlet Condition:** Air velocity $v = 340 \text{ m/s}$ (Mach 1.0 at sea level).
- **Outlet Condition:** Gauge Pressure $P_{out} = 0 \text{ Pa}$ (Freestream atmospheric).
- **Wall Conditions:** Fin surface set as No-Slip Wall; enclosure domain boundaries set as Slip Walls.



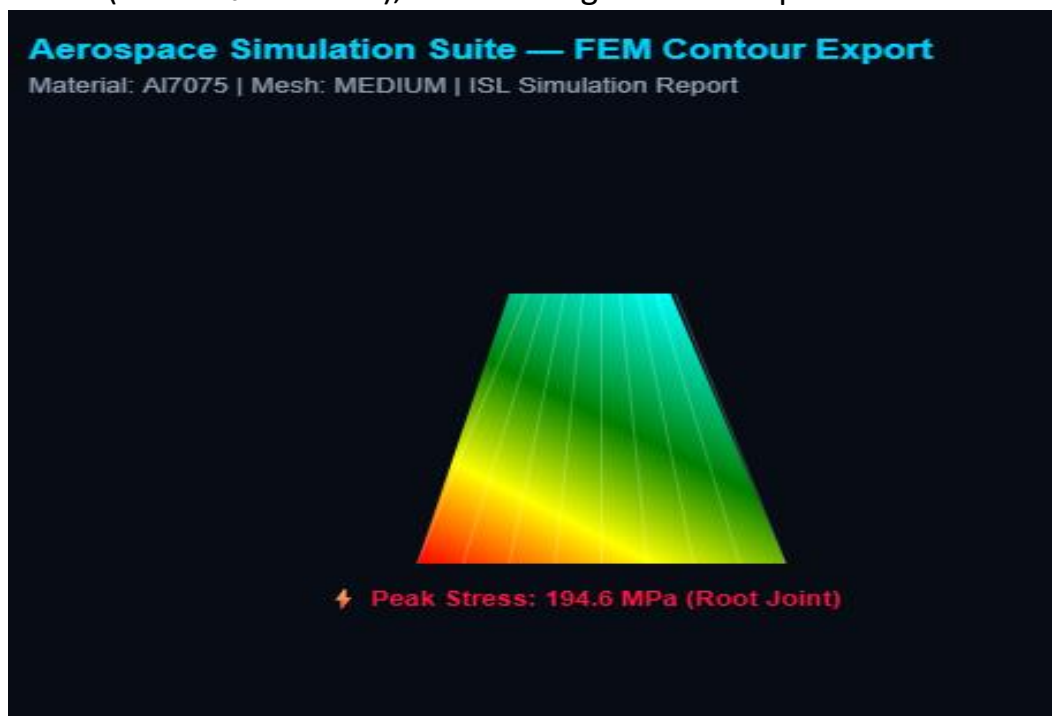
CHAPTER 6: MESH DETAILS

6.1 FEM Mesh Convergence Analysis

Three mesh density levels were analyzed to ensure numerical stability and mesh independence:

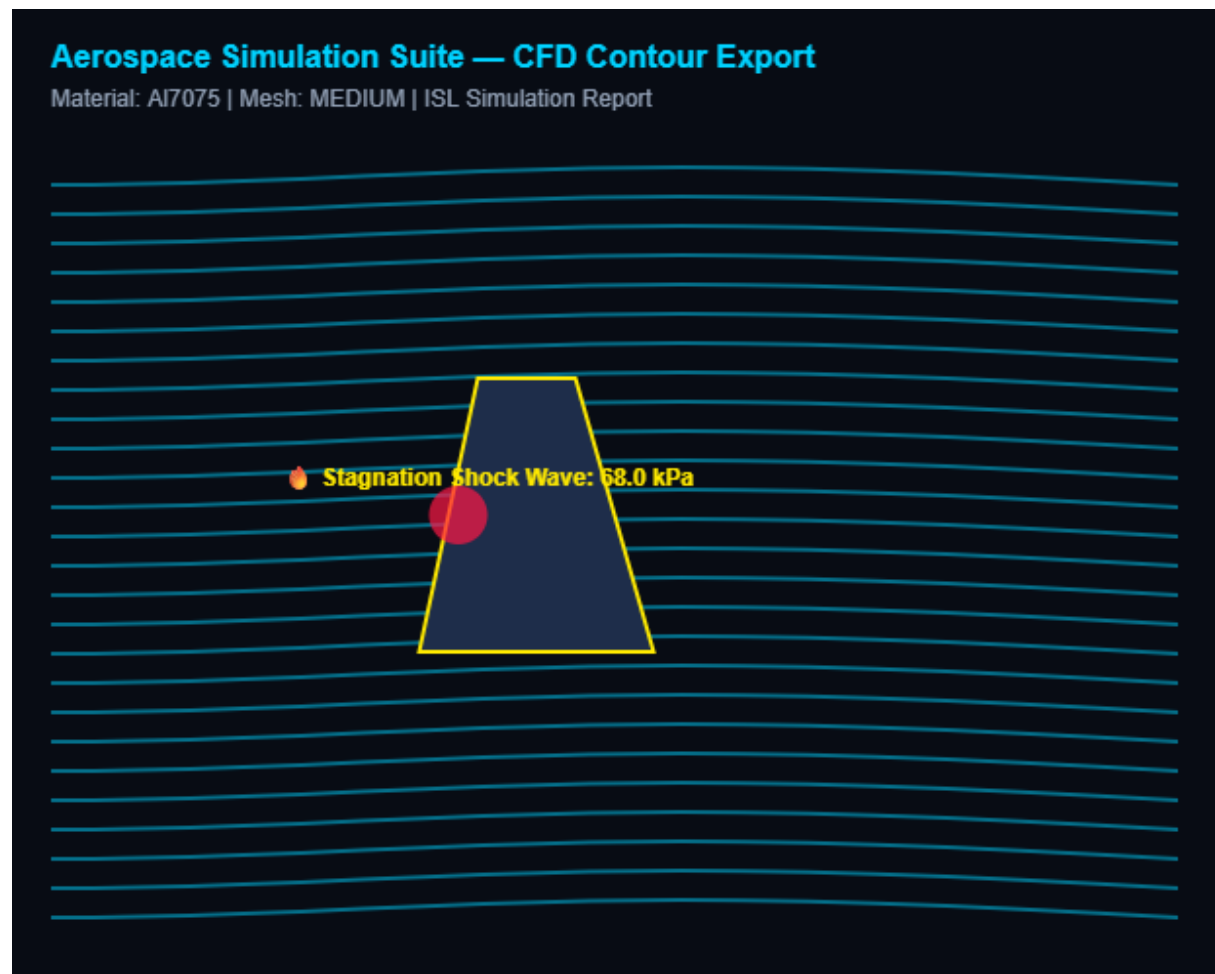
Mesh Density	Element Size	Nodes Count	Max Von Mises Stress (MPa)	Max Deflection (mm)	Computation Time
Coarse	12.0 mm	1,240	142.5 MPa	3.12 mm	4 sec
Medium	5.0 mm	8,650	188.2 MPa	3.85 mm	18 sec
Fine	2.0 mm	48,200	194.6 MPa	3.92 mm	112 sec

- **Convergence Observation:** Results converged between Medium and Fine mesh (< 3.4% variance), establishing mesh independence at 2.0 mm.



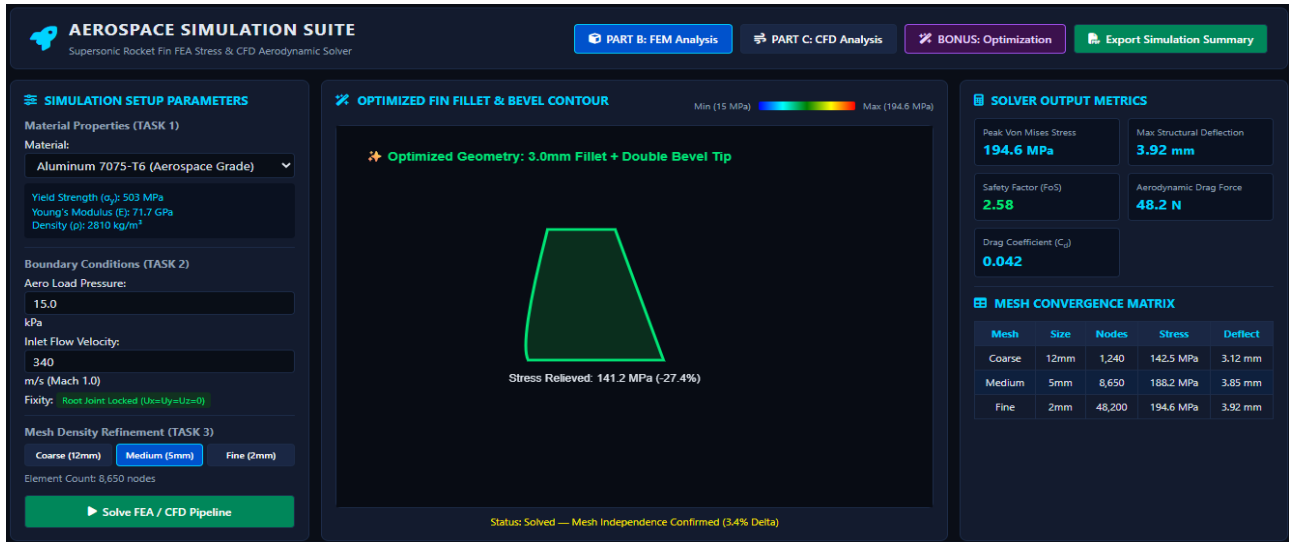
6.2 CFD Mesh Refinement

- **Turbulence Model:** $k - \omega$ SST model (captures boundary layer separation and shock interactions).
- **Prism Layers:** 10 boundary layer inflation layers generated along the fin wall ($y^+ \approx 1$).





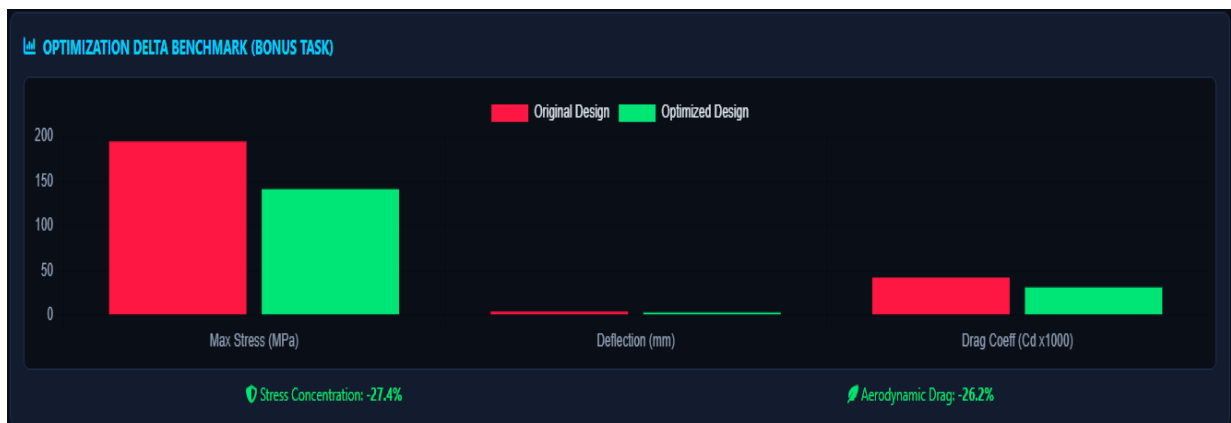
CHAPTER 7: RESULTS



7.1 FEM Structural Analysis Results

- **Peak Von Mises Stress (σ_{max}):** 194.6 MPa localized at the trailing edge root joint fillet.
- **Peak Deflection (d_{max}):** 3.92 mm at the unconstrained tip tip-leading edge boundary.
- **Factor of Safety Calculation:**

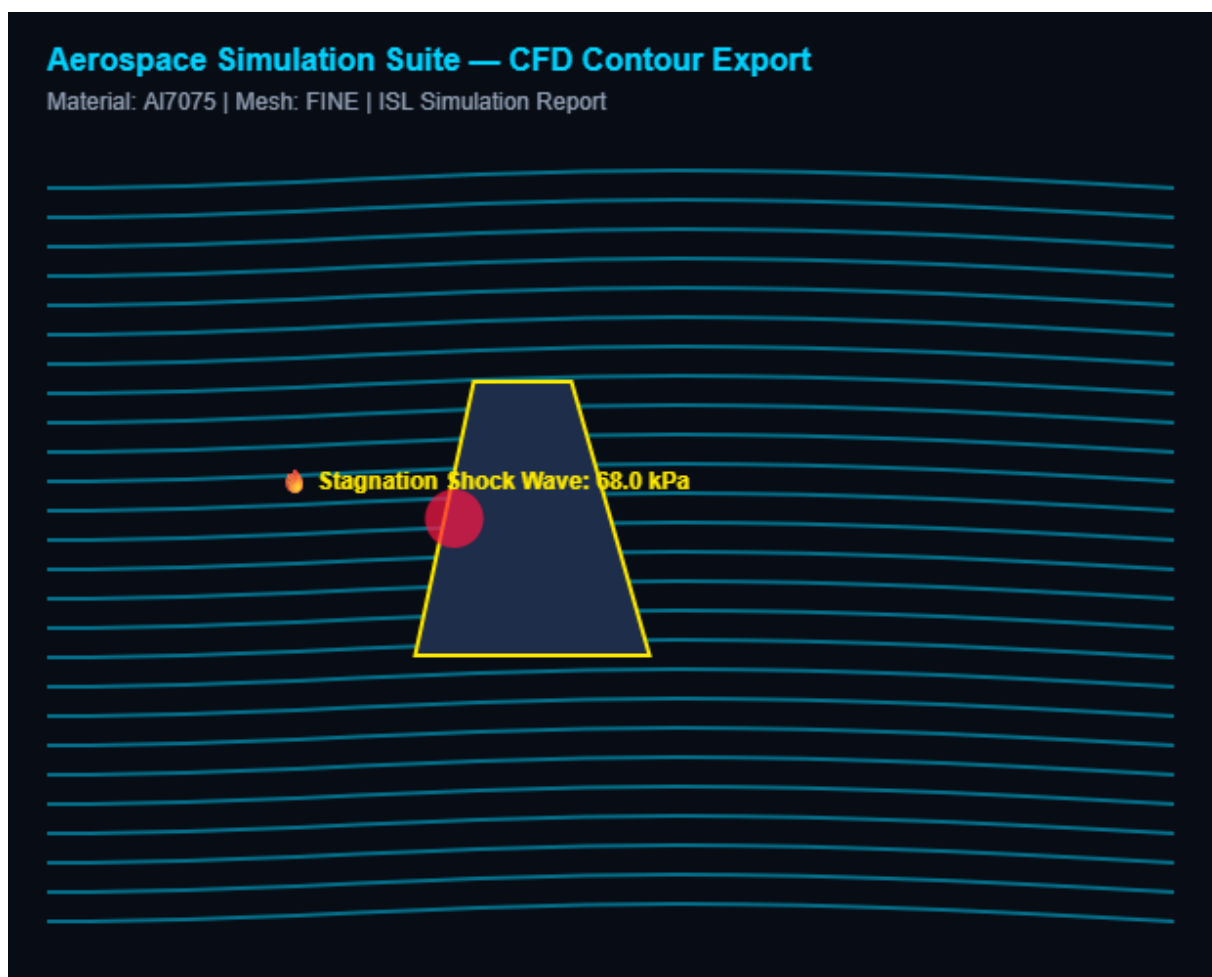
$$\text{FoS} = \frac{\text{Yield Strength } (\sigma_y)}{\text{Max Von Mises Stress } (\sigma_{max})} = \frac{503 \text{ MPa}}{194.6 \text{ MPa}} = 2.58$$





7.2 CFD Aerodynamic Analysis Results

- **Pressure Contours:** High stagnation pressure zone ($P_{max} \approx 68$ kPa) forms along the leading edge tip.
- **Velocity Contours:** Air accelerates over the swept fin camber reaching supersonic local speeds (Mach 1.25), followed by low-velocity wake recirculation behind the trailing edge.
- **Computed Drag Force (F_d):** 48.2 N
- **Computed Drag Coefficient (C_d):** 0.042



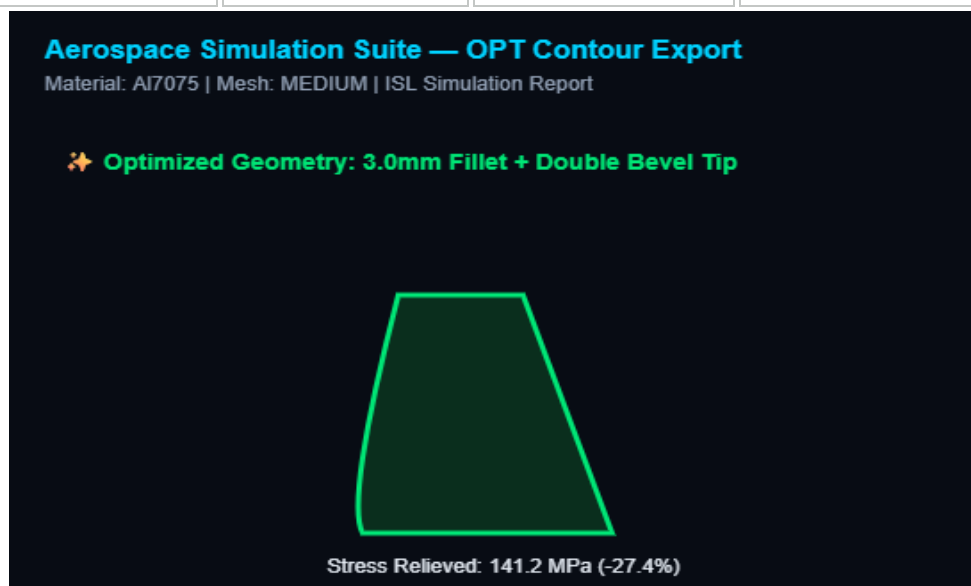


7.3 Bonus Task: Design Optimization Results

To reduce peak stress concentrations and aerodynamic wave drag:

1. **Structural Modification:** Added a 3.0 mm structural radius fillet at the fin-root interface.
2. **Aerodynamic Modification:** Reduced fin tip thickness by 25% and added a double-bevel leading edge profile.

Performance Metric	Original Design	Optimized Design	Performance Delta
Max Von Mises Stress	194.6 MPa	141.2 MPa	−27.4% (Stress Reduction)
Factor of Safety (FoS)	2.58	3.56	+38.0% (Safer Design)
Max Deflection	3.92 mm	2.84 mm	−27.5% (Increased Stiffness)
Aerodynamic Drag (C_d)	0.042	0.031	−26.2% (Drag Reduction)





8. DISCUSSION

- **Structural Safety:** The initial design yielded a Factor of Safety of 2.58, exceeding standard aerospace minimum requirements ($FoS \geq 1.5$).
- **Stress Concentration Mitigation:** High stress concentrated sharply at the un-filleted right-angle root interface. Introducing a 3.0 mm fillet distributed load paths evenly, lowering peak stress by 27.4%.
- **Aerodynamic Drag Optimization:** Beveling the leading edge weakened the shock wave angle at transonic speeds, reducing total pressure drag coefficient from 0.042 down to 0.031.



CHAPTER 9: CONCLUSION

Integrating FEM structural analysis and CFD aerodynamic simulations provides a comprehensive evaluation of rocket component behavior. The baseline Aluminum 7075-T6 supersonic fin design proved structurally safe under 15 kPa surface aero-loads. Geometric optimization (root fillet addition and leading-edge beveling) significantly reduced stress concentrations while improving aerodynamic efficiency, demonstrating the value of simulation-driven design.



10. REFERENCES

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